# LABSHEET-09

**Q1. Display the mounted file systems and their disk-space usage using df.**

df

**Q2. Display disk-space usage in human-readable format using df and identify the filesystem, total, used, available and mounted-on fields.**

df -h

**Q3. Display the size of the current directory and its contents using du.**

du

**Q4. Display directory sizes in human-readable format and identify which directories are consuming the most space.**

du -h --max-depth=1 | sort -hr

**Q5. Display the size of a selected directory including its subdirectories using du with an appropriate option.**

du -sh /var/log/*

**Q6. Display all available block devices, disks and partitions using lsblk.**

lsblk

**Q7. Use lsblk to identify the disk names, partition names, sizes, types and mount points on the system.**

lsblk -o NAME,SIZE,TYPE,MOUNTPOINT

**Q8. Display additional file-system information for block devices using an appropriate lsblk option.**

lsblk -f

**Q9. Identify the currently mounted file systems and their mount points using findmnt or an appropriate command.**

findmnt

**Q10. Create a dedicated directory named /mnt/labdisk to use as a mount point and verify its permissions and existence.**

sudo mkdir -p /mnt/labdisk
ls -ld /mnt/labdisk

**Q11. Attach a suitable test storage device or partition to /mnt/labdisk and verify that it is mounted correctly.**

sudo mount /dev/sdb1 /mnt/labdisk
findmnt /mnt/labdisk

**Q12. Display the disk-space usage of the mounted /mnt/labdisk file system using df.**

df -h /mnt/labdisk

**Q13. Create a test file inside /mnt/labdisk and verify that the file is stored on the mounted file system.**

echo 'Test file content' | sudo tee /mnt/labdisk/testfile.txt
ls -l /mnt/labdisk

**Q14. Unmount /mnt/labdisk using umount and verify that the file system is no longer mounted.**

```
sudo umount /mnt/labdisk
findmnt /mnt/labdisk
```

**Q15. Attempt to access a mounted file system that is currently in use, identify why unmounting may fail, and safely release the process before unmounting.**

```
# Identify processes using the mount point:
sudo fuser -v /mnt/labdisk
# Kill active processes or exit the directory, then unmount:
sudo fuser -k /mnt/labdisk
sudo umount /mnt/labdisk
```

**Q16. Display the partition table of a selected disk using an appropriate command and identify the partitions and their sizes.**

```
sudo fdisk -l /dev/sdb
```

**Q17. Create a practice partition on a dedicated test disk or virtual disk using an appropriate partitioning tool, then verify it with lsblk. Do not modify the system disk.**

```
sudo fdisk /dev/sdb
# Press 'n' to create partition, 'p' for primary, set size, then 'w' to save
lsblk /dev/sdb
```

**Q18. Create a file system on a dedicated practice partition, mount it on /mnt/labdisk and verify the file system using df, lsblk and findmnt.**

```
sudo mkfs.ext4 /dev/sdb1
sudo mount /dev/sdb1 /mnt/labdisk
df -h /mnt/labdisk
lsblk /dev/sdb1
findmnt /mnt/labdisk
```

**Q19. Unmount the practice file system and verify the mount point, device and partition information after unmounting.**

```
sudo umount /mnt/labdisk
findmnt /mnt/labfs
lsblk /dev/sdb1
```

**Q20. Perform a complete disk and storage management workflow: inspect disks and partitions, check disk usage, mount a dedicated test partition, create and verify data, unmount it and confirm the final storage state.**

```
lsblk
df -h
sudo mkdir -p /mnt/labdisk
sudo mount /dev/sdb1 /mnt/labdisk
echo 'Disk Workflow Test' | sudo tee /mnt/labdisk/data.txt
ls -l /mnt/labdisk
sudo umount /mnt/labdisk
findmnt /mnt/labdisk
```